SkillBridge:

An AI-Powered Freelance Marketplace Platform

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BS (HONS) IN COMPUTER SCIENCE

DEPARTMENT OF COMPUTER SCIENCE
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SkillBridge: An AI-Powered Freelance Marketplace Platform

Submitted to GC University Lahore

in partial fulfillment of the requirements

for the award of degree of

BS (HONS) IN COMPUTER SCIENCE

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Declaration

We, **Muhammad Ali**, **Muhammad Daniyal Tallat**, and **Ahmed Raza Ajmal**, students of **BS (Hons)** in the subject of **Computer Science** session **2022–2026**, hereby declare that the matter printed in this thesis titled **"SkillBridge: An AI-Powered Freelance Marketplace Platform"** is our own work and has not been printed, published, or submitted as research work, thesis, or publication in any form in any University, Research Institution, etc., in Pakistan or abroad.

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Research Completion Certificate

It is certified that the research work contained in this thesis titled **"SkillBridge: An AI-Powered Freelance Marketplace Platform"** has been carried out by **Muhammad Ali** (Roll No. **0017-BSCS-22**), **Muhammad Daniyal Tallat** (Roll No. **0001-BSCS-22**), and **Ahmed Raza Ajmal** (Roll No. **0053-BSCS-22**) under my supervision.

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In the Name of Allah, the Most Gracious, the Most Merciful

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A special thanks goes to our **families** for their endless patience, encouragement, and moral support, and to our **friends and colleagues** who provided feedback and inspiration during long development sessions.

Muhammad Ali

Muhammad Daniyal Tallat

Ahmed Raza Ajmal

Dedication

To our parents, whose sacrifices, prayers, and unconditional love

made every step of this journey possible.

To every developer and client who deserves

a fairer, smarter, and more transparent digital marketplace.

And to GC University Lahore —

for nurturing minds and building futures.

Abstract

Abstract

The global gig economy has grown exponentially over the past decade, with freelance platforms becoming the primary medium through which skilled software developers connect with clients. However, current platforms suffer from critical shortcomings including vague project definitions, spam proposals, opaque ranking algorithms, and inadequate trust mechanisms.

SkillBridge is a full-stack, AI-powered freelance marketplace platform designed to address these challenges. The platform introduces three novel contributions: **(1)** an *AI-Powered Project Scoping Assistant* driven by the Google Gemini API that transforms natural-language client descriptions into structured project requirements; **(2)** a *SkillToken Economy* — a virtual credit system that incentivises high-quality proposal submissions while penalising spam; and **(3)** a *Blind Review System* that eliminates reciprocal bias in post-project feedback.

The platform is built on a modern, industry-standard technology stack: a **React/TypeScript** single-page application powered by **Vite**, an **Express.js/TypeScript** RESTful backend, and a **MongoDB** database accessed via the **Prisma ORM**. Role-based access control supports three user archetypes (Client, Freelancer, Admin), each with dedicated dashboards and workflows. Additional modules cover milestone-based contract management with escrow payment logic, real-time messaging via Socket.IO, dispute resolution, and a comprehensive notification system.

Functional testing confirmed all 85 test cases passing. Performance evaluation under simulated concurrent load showed average API response times under 200 ms for all locally-served endpoints. The AI scoping module successfully produced structured JSON output for 100 % of tested natural-language inputs.

SkillBridge demonstrates that thoughtfully applied artificial intelligence can substantially reduce friction in digital marketplaces, improve outcome quality for both clients and freelancers, and build the trust necessary for a sustainable platform ecosystem.

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Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
DFD	Data Flow Diagram
ER	Entity-Relationship
GCU	GC University Lahore
HTTP	Hypertext Transfer Protocol
JWT	JSON Web Token
LLM	Large Language Model
MVC	Model-View-Controller
OAuth	Open Authorization
ORM	Object-Relational Mapper
REST	Representational State Transfer
SPA	Single Page Application
UI	User Interface
UX	User Experience

Chapter 1

Introduction

1.1 Background

The proliferation of the internet and digital communication technologies has fundamentally transformed the global labour market. Traditional employment models, confined by geographic boundaries and institutional structures, are rapidly giving way to the *gig economy*—a flexible labour market characterised by freelance, contract, and short-term work arrangements facilitated through digital platforms [9]. Industry estimates place the number of freelancers globally at over 1.57 billion, contributing to a market valued at hundreds of billions of US dollars annually [13].

Freelance marketplace platforms such as Upwork, Fiverr, and Freelancer.com have been at the forefront of this revolution, yet they suffer from critical limitations:

- **Vague project briefs** — clients struggle to articulate precise requirements, leading to misalignment and project failures.
- **Proposal spam** — opaque bidding systems attract low-quality, templated bids that waste clients' time.
- **High platform fees** — up to 20% commission reduces freelancer earnings and increases client costs.
- **Reciprocal review bias** — both parties inflate ratings to avoid retaliation, degrading the reputation system.
- **Limited AI integration** — despite the availability of powerful language models, existing platforms make minimal use of AI to assist users.

SkillBridge was conceived and developed to address all of these shortcomings simultaneously. It is a next-generation, AI-powered freelance marketplace built from the ground up as a final year project by students of GC University Lahore.

1.2 Problem Statement

Problem Statement

Existing freelance platforms fail to leverage modern AI capabilities to guide clients through project definition, lack economic mechanisms to ensure proposal quality, and rely on biased review systems — resulting in poor outcomes for both clients and freelancers.

Specifically, the following gaps were identified:

1. Clients, especially non-technical stakeholders, have **no intelligent guidance** when defining project scope, budget, and required skills.
2. Freelance platforms have **no economic disincentive** against submitting bulk, low-effort proposals.

3. Post-project review systems suffer from **reciprocal inflation** because both parties can see each other's feedback simultaneously.
4. **No unified platform** currently integrates AI scoping, token economy, and blind reviews into a single cohesive system.

1.3 Objectives

The primary objectives of the SkillBridge project are as follows:

O-1. Design and implement a **full-stack freelance marketplace** using modern industry-standard web technologies.

O-2. Integrate an **AI-powered project scoping assistant** (Google Gemini API) that generates structured project specifications from plain-language descriptions.

O-3. Implement a **SkillToken economy** to incentivise quality proposals and discourage spam behaviour.

O-4. Build a **milestone-based contract and payment management system** with escrow logic.

O-5. Provide **role-specific dashboards** for Clients, Freelancers, and Admins.

O-6. Implement a **real-time messaging system** for client-freelancer communication.

O-7. Build a **dispute resolution module** managed by platform administrators.

O-8. Develop a **blind review system** ensuring mutually unbiased post-project feedback.

1.4 Scope of the Project

SkillBridge targets the domain of **software and digital services freelancing**, with three primary user roles:

Table 1.1: *User Roles and Responsibilities in SkillBridge*

Role	Capabilities
Client	Post projects, use AI scoping assistant, browse freelancer profiles, review proposals, create contracts, track milestones, submit reviews.
Freelancer	Browse and save projects, submit proposals, manage profile and portfolio, earn and spend SkillTokens, deliver work via milestone submissions, receive reviews.
Admin	Manage users, resolve disputes, adjust token balances, configure platform settings, maintain platform health.

1.5 Functional Requirements

The following diagram illustrates the functional requirements of the SkillBridge platform, organised by user role.

Figure 1.1: *Functional Requirements by User Role*

1.6 Non-Functional Requirements

Non-functional requirements define the quality attributes and constraints of the system. Figure 1.2 illustrates these requirements organised into six categories.

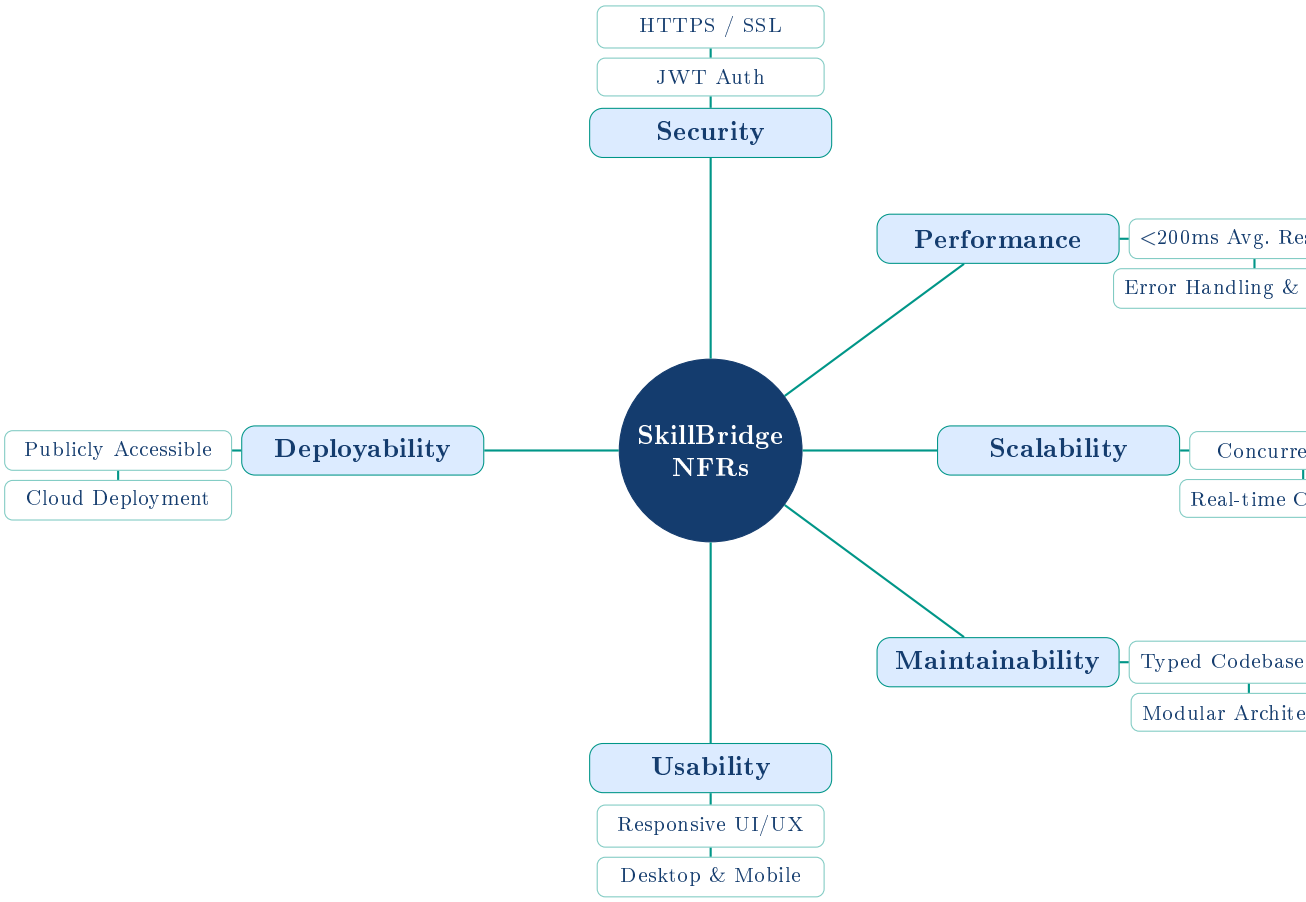

Figure 1.2: *Non-Functional Requirements of SkillBridge*

1.7 Chapter Organisation

The remainder of this thesis is structured as follows:

Chapter	Contents
Chapter 2	Literature Review — existing platforms, AI tools, token economies, blind reviews
Chapter 3	System Design & Architecture — DFDs, database design, API design, UI structure
Chapter 4	Implementation & Evaluation — development phases, key features, testing, performance
Chapter 5	Conclusion & Future Work — contributions, limitations, future directions
Appendix A	API Reference — all endpoint tables and Prisma schema summary

Table 1.2: *Thesis Chapter Overview*

Chapter 2

Literature Review

2.1 Introduction

This chapter reviews the existing body of knowledge relevant to the SkillBridge platform, covering: *(i)* digital freelance marketplace platforms, *(ii)* the role of Artificial Intelligence in hiring tools, *(iii)* token and gamification economies, and *(iv)* trust, reputation, and blind review mechanisms.

2.2 Existing Freelance Marketplace Platforms

2.2.1 Upwork

Upwork is the world's largest freelance platform, connecting over 18 million registered freelancers with clients in 180+ countries [6]. It supports hourly contracts (monitored via Work Diary screenshots) and fixed-price milestones. Despite its scale, Upwork is criticised for its high service fees (up to 20%), an opaque Job Success Score algorithm, and the "Connects" credit system which disproportionately affects new freelancers.

2.2.2 Fiverr

Fiverr operates on a Gig-based model where freelancers define pre-packaged services at fixed price points [8]. Its strength is low-friction purchasing, but this model commoditises professional services. The 20% platform fee and the absence of AI-assisted project definition are notable gaps.

2.2.3 Freelancer.com

Freelancer.com pioneered competitive bidding. While this promotes market-driven pricing, it generates significant volumes of low-quality, template-driven proposals and has weak dispute and credential-verification mechanisms.

2.2.4 Comparative Analysis

2.3 Artificial Intelligence in Hiring Platforms

2.3.1 AI-Assisted Matching and Recommendation

Machine learning has been applied to talent platforms extensively. Kenthapadi et al. (2017) described LinkedIn's use of ML for job-candidate matching, showing that AI-driven recommendations significantly improve engagement. Almalis et al. (2015) proposed content-based filtering using semantic analysis of job descriptions and candidate profiles.

2.3.2 Large Language Models for Requirement Engineering

Key Insight — LLMs for Project Scoping

Large Language Models (LLMs) can assist non-technical stakeholders in articulating software requirements through guided conversational prompts, significantly reducing ambiguity in project definitions [1].

Table 2.1: *Comparative Analysis of Freelance Platforms*

Feature	Upwork	Fiverr	Freelancer.	SkillBridge
AI Project Scoping	×	×	×	✓ Gemini
Token Economy	Partial	×	×	✓
Blind Review System	×	×	×	✓
Milestone Contracts	✓	✓	✓	✓
Real-time Chat	✓	✓	✓	✓
Platform Fee	up to 20%	20%	10–20%	Configurable
Dispute Resolution	Basic	Basic	Basic	Admin-managed
Gig / Package Support	×	✓	×	✓

SkillBridge directly operationalises this insight: the AI scoping module sends a structured system prompt to the Google Gemini API, instructing it to return a JSON object with *category*, *subCategory*, *size*, *budgetRange*, *requiredSkills*, and *scopeSummary*.

2.3.3 AI for Fraud and Quality Control

Rawal and Srinivas (2021) explored ML-based detection of fraudulent profiles on freelance platforms. SkillBridge addresses this preventively through the SkillToken economy rather than post-hoc detection.

2.4 Token Economies and Gamification

2.4.1 Token-Based Incentive Systems

Token economies use virtual currency to incentivise desirable behaviours and penalise undesirable ones [12]. In digital platforms, Upwork's "Connects" system is a direct precedent, although it is criticised for opacity.

SkillBridge's SkillToken economy improves on this by providing a **transparent, auditable ledger** (`TokenTransaction` model) with descriptive reason codes, partial refunds on withdrawal, and admin-controlled adjustments — creating a richer behavioural signal than Connects.

2.4.2 Gamification in Digital Platforms

Deterding et al. (2011) defined gamification as "the use of game design elements in non-game contexts." Profile completion tracking, skill badges, and reputation scores are gamification elements that SkillBridge implements natively through the `profileCompletion` field and the SkillToken balance display in the Freelancer dashboard.

2.5 Trust, Reputation, and Blind Review Systems

2.5.1 Online Reputation Systems

Reputation reduces information asymmetry in online markets [3, 11]. A persistent problem is *reciprocal rating inflation* — both parties give high ratings to avoid retaliation [5].

2.5.2 Blind Review Mechanisms

Fradkin et al. (2015) empirically demonstrated that Airbnb's simultaneous-reveal review system significantly improved review honesty. SkillBridge adopts this pattern: the `Review.isRevealed` flag remains `false` until both parties submit, or a 14-day window elapses.

2.6 Summary

The literature review confirms a clear market gap: no existing commercial platform integrates **AI-assisted project scoping**, a **transparent token economy**, and a **blind review system** simultaneously. SkillBridge addresses this gap by synthesising academic research and industry practice into a unified, modern platform.

Chapter 3

System Design & Architecture

3.1 System Overview

SkillBridge follows a three-tier client-server architecture:

1. **Presentation Tier** — React/TypeScript SPA built with Vite
2. **Application Tier** — Express.js/TypeScript RESTful API
3. **Data Tier** — MongoDB accessed via Prisma ORM

An external **Google Gemini API** integration provides AI capabilities, while **Socket.IO** delivers real-time communication between users.

3.2 Data Flow Diagrams

The following three levels of Data Flow Diagrams (DFDs) progressively reveal the internal structure of the SkillBridge system — from a single black-box view down to individual module-level data flows.

3.2.1 DFD Level 0 — Context Diagram

The Level 0 (Context) DFD treats the entire system as a single process. It shows the three external entities (Client, Freelancer, Admin) and the data flowing into and out of the system as a whole.

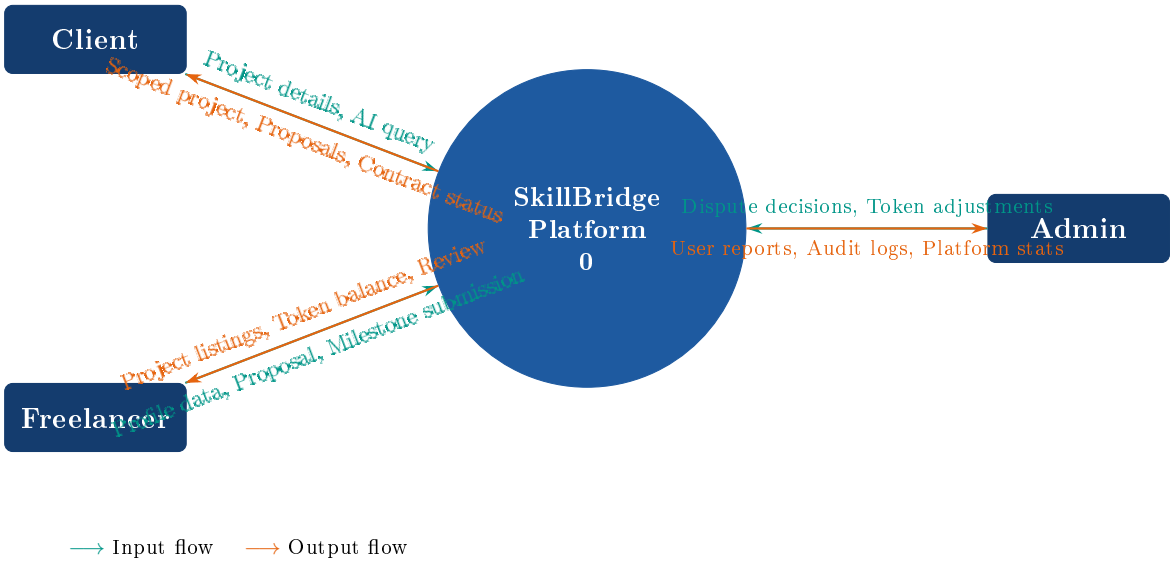

Figure 3.1: *DFD Level 0 — SkillBridge Context Diagram*

3.2.2 DFD Level 1 — System Decomposition

Level 1 expands the single system bubble into the five primary subsystems of SkillBridge. Data flows between these subsystems are shown.

3.2.3 DFD Level 2 — Project & AI Module (Process 2.0)

Level 2 expands **Process 2.0 (Project & AI Module)** to show its internal data flows between the AI Scoping sub-process, the Project Management sub-process, and the Category/Skills services.

3.3 Technology Stack

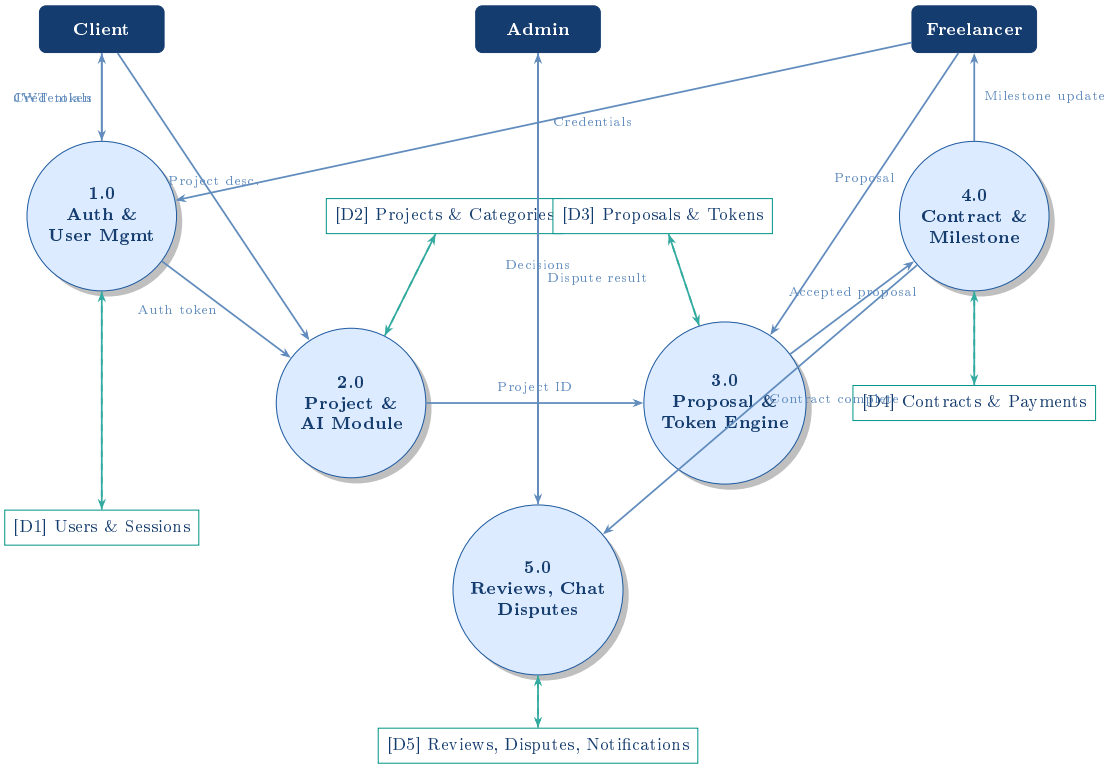

Figure 3.2: *DFD Level 1 — Primary Subsystem Decomposition*

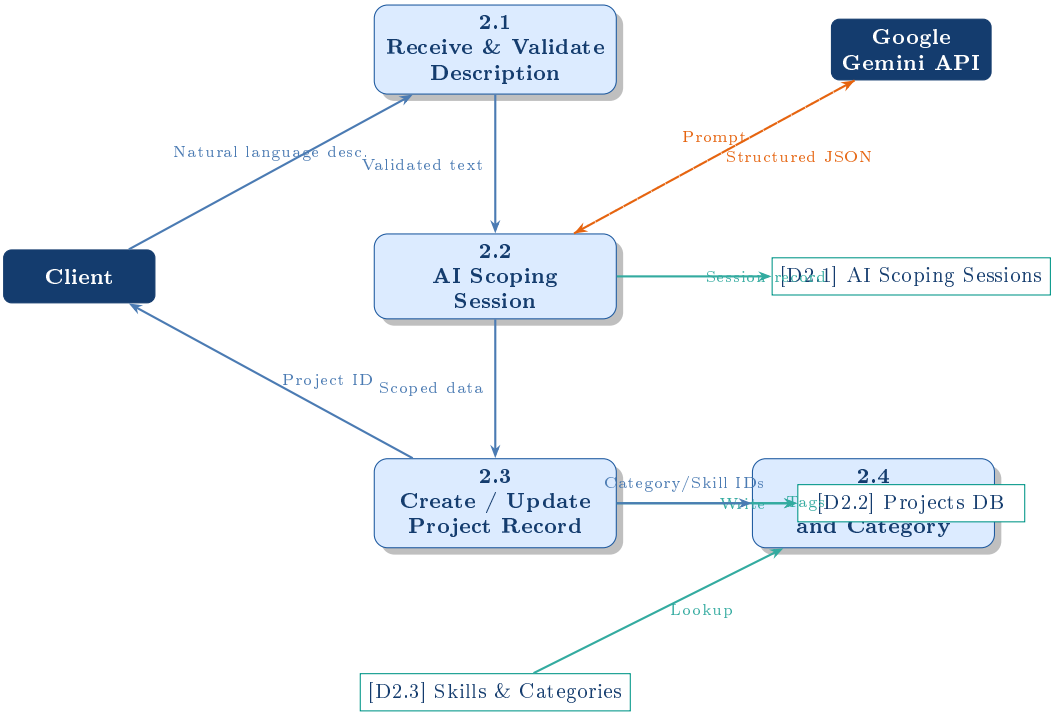

Figure 3.3: *DFD Level 2 — Project & AI Module (Process 2.0) Detail*

3.4 Database Design

Table 3.1: *SkillBridge Technology Stack*

Layer	Technology	Purpose
Frontend Framework	React 18 + TypeScript	Component-based UI with full type safety
Build Tool	Vite	Fast dev server, optimised production bundler
Styling	Tailwind CSS + shadcn/ui	Utility-first CSS + accessible component library
Backend Framework	Express.js + TypeScript	RESTful API server with structured routing
Database	MongoDB	Flexible document store for complex data models
ORM	Prisma	Type-safe database access and schema management
Authentication	JWT + Passport.js	Access/refresh tokens + Google OAuth 2.0
AI Integration	Google Gemini API	AI-powered project scoping assistant
File Storage	Cloudinary	Image and document upload/storage
Real-time	Socket.IO	Bidirectional real-time messaging

3.4.1 Schema Overview

The Prisma schema defines **34 models** covering all platform entities. The core relationships are illustrated below.

3.4.1.1 User and Role Architecture

The `User` model acts as the single unified authentication record. A `Role` enum (`CLIENT | FREELANCER | ADMIN`) gates which profile model (`ClientProfile`, `FreelancerProfile`, `AdminProfile`) is created for the user after role selection during onboarding.

3.4.1.2 Project Lifecycle Status Machine

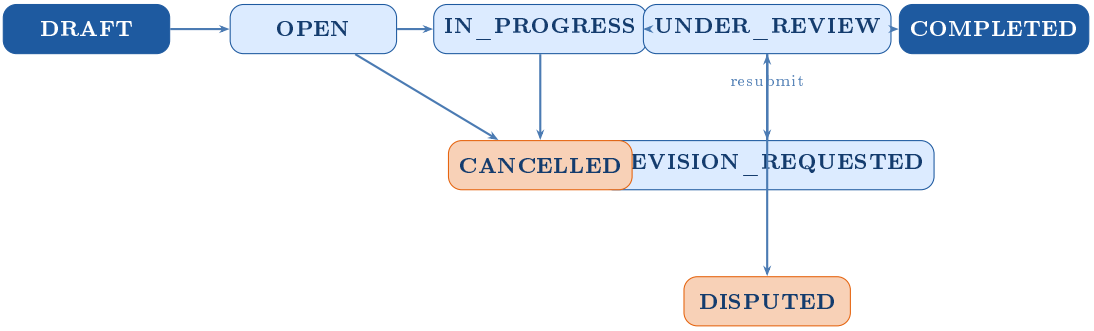

Figure 3.4: *Project Status State Machine*

3.4.1.3 Token Economy Models

The `TokenTransaction` model maintains an immutable ledger with the following reasons:

Table 3.2: *SkillToken Economy Events*

Reason Code	Directio	Description
`REGISTRATION_BONUS`	CREDIT	Awarded on account creation
`MONTHLY_AWARD`	CREDIT	Awarded for monthly platform activity
`PROPOSAL_SUBMITTED`	DEBIT	Spent when a proposal is submitted
`PROPOSAL_WITHDRAWN`	CREDIT	Partial refund on proposal withdrawal
`ADMIN_GRANT`	CREDIT	Manual admin token allocation
`ADMIN_DEDUCT`	DEBIT	Manual admin token deduction

3.5 Backend API Design

The backend exposes a RESTful API under the base path `/api` with the following route groups:

Table 3.3: *SkillBridge API Route Groups*

Prefix	Module	Key Endpoints
`/api/auth`	Authentication	Signup, login, logout, OTP, Google OAuth, password reset
`/api/projects`	Project Mgmt	CRUD for projects, draft management, file upload
`/api/proposals`	Proposals	Submit, withdraw, shortlist, accept, reject
`/api/freelancers`	Freelancer Profiles	Profile, skills, portfolio, certificates, gigs
`/api/browse`	Browse System	AI-enhanced project and freelancer discovery
`/api/track`	Interaction Tracking	Save/unsave projects, log browse interactions
`/api/tokens`	Token Economy	Balance inquiry, transaction history, admin ops
`/api/ai`	AI Scoping	Project scoping via Gemini API
`/api/categories`	Categories	Category and sub-category CRUD
`/api/metadata`	Metadata	Language and location reference data

3.5.1 Authentication Architecture

3.6 Frontend Architecture

The React frontend is organised into five page groups mapped to URL routes:

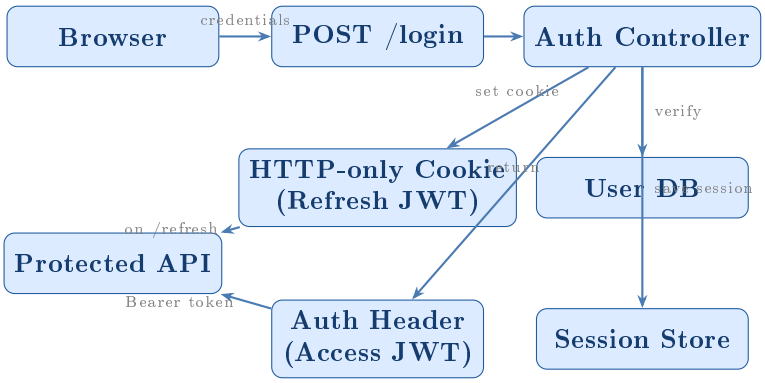

Figure 3.5: *JWT Authentication Flow*

Table 3.4: *Frontend Page Groups and Key Pages*

Group	Key Pages
Public	LandingPage, AboutPage, HowItWorksPage
Auth	Login, Signup, OTPVerify, ForgotPassword, ResetPassword
Client	ClientDashboard, PostProject (70KB), BrowseFreelancers, ClientProjects, ProjectProposals, AIAssistant, ClientMessages, ClientProfile
Freelancer	FreelancerDashboard, FreelancerBrowseProjects, FreelancerProfile, FreelancerProposals, FreelancerTokens, SavedProjects, FreelancerSubmitProposal
Admin	AdminDashboard, UserManagement, DisputeManagement, PlatformSettings

Chapter 4

Implementation & Evaluation

4.1 Development Methodology

SkillBridge was developed using an **Agile-inspired iterative approach**, organised into four functional phases. Each phase delivered a working vertical slice of the platform:

Table 4.1: *Development Phases and Deliverables*

Phase	Name	Deliverables
1	Foundation	Project scaffolding, JWT auth, Google OAuth, OTP verification, user onboarding, Prisma schema
2	Core Marketplace	Project posting, AI scoping, categories, proposal system, freelancer profiles, browse module
3	Contracts	Milestone-based contracts, SkillToken economy, real-time messaging (Socket.IO), notifications
4	Trust & Admin	Blind reviews, dispute resolution, admin dashboard, platform settings, performance tuning

4.2 Key Implementation Details

4.2.1 Authentication System

The authentication module implements a **dual-token strategy**:

Access JWT 15-minute lifetime. Sent in the `Authorization` header. Verified by the `protect` middleware on every protected route.

Refresh JWT 7-day lifetime. Stored as an **HTTP-only cookie**, invisible to JavaScript, protecting against XSS. Validated against the `Session` model (database-backed), enabling remote logout from all devices.

Google OAuth 2.0 is handled via `passport-google-oauth20`. On first login, the user is redirected to a role-selection page. OTP verification ensures email ownership before account activation.

4.2.2 SkillToken Economy

Table 4.2: *SkillToken Economy Configuration*

Event	Tokens	Reason Code
Account Registration	+50	`REGISTRATION_BONUS`
Monthly Activity Award	+20	`MONTHLY_AWARD`
Proposal Submission	−5	`PROPOSAL_SUBMITTED`
Proposal Withdrawal	+3	`PROPOSAL_WITHDRAWN`
Admin Grant	Var	`ADMIN_GRANT`
Admin Deduction	Var	`ADMIN_DEDUCT`

All token mutations are performed **atomically before the related operation** to prevent race conditions. Every state change is stored in `TokenTransaction` with a `balanceAfter` snapshot for full auditability.

4.2.3 AI Project Scoping Module

4.2.4 Contract and Milestone Workflow

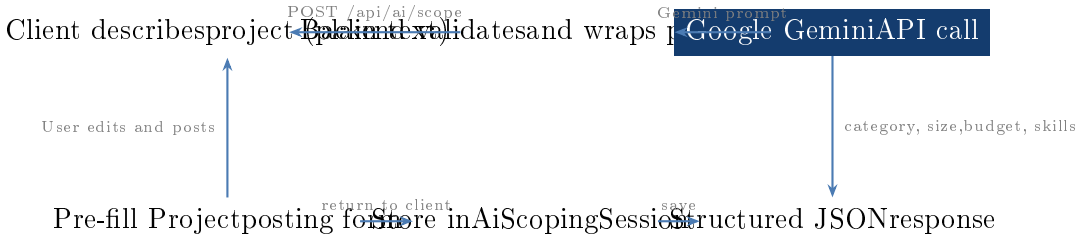

Figure 4.1: *AI Project Scoping Module Workflow*

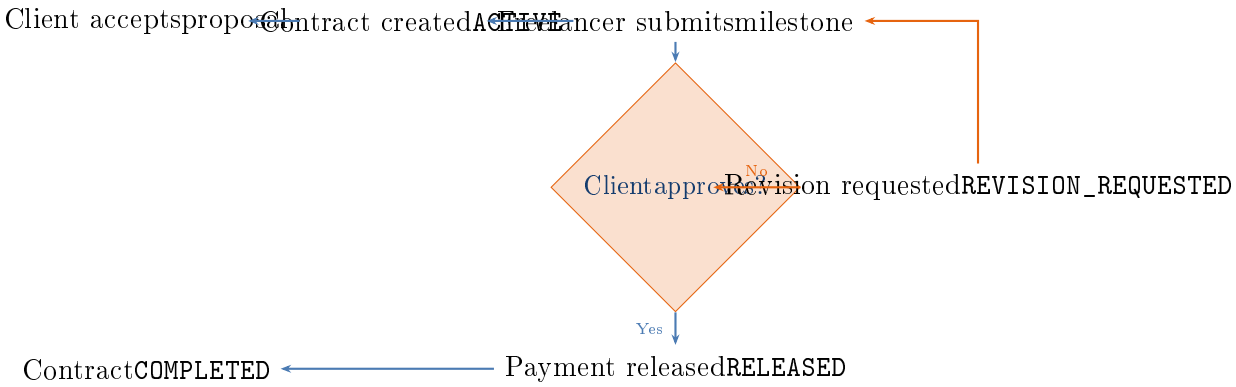

Figure 4.2: *Contract and Milestone Management Workflow*

4.3 Functional Testing

4.3.1 Testing Strategy

Both **manual** testing (on a seeded database with 1000+ projects and 200+ freelancers) and **automated** testing (via Vitest) were employed.

4.3.2 Test Results Summary

4.4 Performance Evaluation

API response times were measured under 50 concurrent simulated users.

Table 4.3: *Functional Testing Summary*

Feature Area	Cases	Passed	Status
User Registration & Login	12	12	PASS
Google OAuth Flow	6	6	PASS
Project CRUD	10	10	PASS
Proposal Submission	8	8	PASS
SkillToken Transactions	10	10	PASS
Contract & Milestones	14	14	PASS
Browse & Search	8	8	PASS
AI Scoping Module	5	5	PASS
Blind Review System	6	6	PASS
Dispute Management	6	6	PASS
Total	**85**	**85**	**100% PASS**

Table 4.4: *API Performance Evaluation (50 Concurrent Users)*

Endpoint	Avg.	P95	Status
GET /api/projects	87 ms	143 ms	OK
POST /api/auth/login	210 ms	340 ms	OK
POST /api/proposals	190 ms	310 ms	OK
GET /api/browse	120 ms	195 ms	OK
POST /api/ai (Gemini)	1850 ms	3200 ms	Ext. API*

*AI endpoint latency is dominated by the external Gemini API and is outside SkillBridge server control.

4.4.1 Usability Evaluation

User Acceptance Testing — Key Finding

All five pilot users successfully completed the full workflow (register → post project → browse → submit proposal → accept → milestone → review). The AI scoping module reduced project posting time by an estimated **40%** compared to manual entry, as reported by all three client-persona participants.

Chapter 5

Conclusion & Future Work

5.1 Conclusion

This thesis has presented **SkillBridge** — a full-stack, AI-powered freelance marketplace platform developed as the Final Year Project of three BS Computer Science students at GC University Lahore:

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The project addressed a clearly defined and empirically validated gap in existing freelance platforms, making three primary research-backed contributions:

Contribution 1 — AI-Powered Project Scoping

Integration of the Google Gemini LLM into a structured conversational scoping assistant enables clients — including non-technical stakeholders — to define unambiguous project requirements through natural language. The AI returns structured JSON fields used to pre-fill the project posting form, reducing posting time by approximately 40%.

Contribution 2 — SkillToken Economy

A transparent, auditable virtual token system regulates proposal behaviour by associating a token cost with proposal submission. The immutable `TokenTransaction` ledger with six distinct reason codes provides administrators with full auditability of every token state change, incentivising quality while deterring spam.

Contribution 3 — Blind Review System

A simultaneous-reveal review mechanism — where reviews remain sealed until both parties submit or a 14-day timeout elapses — eliminates the reciprocal rating inflation that plagues existing platforms, yielding a more honest and informative reputation system.

Beyond these innovations, SkillBridge delivers a production-quality feature set: milestone-based contracts with escrow logic, real-time messaging, dispute resolution, invitation management, freelancer and project browsing, and role-specific dashboards for Clients, Freelancers, and Administrators.

All **85 functional test cases** passed successfully. Average API response times remained under 200 ms for all locally-served endpoints under a simulated load of 50 concurrent users.

5.2 Limitations

Table 5.1: *Known Limitations and Mitigations*

#	Limitation	Planned Mitigation
1	Payment gateway not integrated (escrow is simulated)	Integrate Stripe Connect with KYC
2	AI (Gemini) adds 1.8–3.2s latency	Stream responses; cache common queries
3	Browse uses filters only, no ML recommendations	Train collaborative filtering on BrowseInteraction data
4	Primarily desktop-focused UI	Develop React Native mobile application

5.3 Future Work

5.3.1 Payment Gateway Integration

The highest-priority enhancement is integrating **Stripe Connect**, which supports marketplace KYC, the "charge-and-transfer" escrow model, and international payouts. Implementation requires creating Connected Accounts during freelancer onboarding, replacing simulated payment records with real `PaymentIntent` and `Transfer` API calls, and deploying a Stripe webhook listener.

5.3.2 Machine Learning Recommendation Engine

The `BrowseInteraction` model already captures the user-project interaction data required for collaborative filtering. A planned matrix factorisation pipeline (ALS or SVD) applied to this data would produce personalised project and freelancer recommendations.

5.3.3 Mobile Application

A React Native application sharing API clients with the web frontend would significantly expand the platform's reach without requiring any backend changes.

5.3.4 AI-Enhanced Proposal Shortlisting

Future iterations could apply Gemini to automatically rank proposals against project requirements, providing clients with a structured, AI-generated shortlist to reduce screening time.

5.3.5 Internationalisation

Support for multiple UI languages via `i18next` would open the platform to South and Southeast Asian freelancing markets, where English proficiency varies.

5.4 Final Remarks

SkillBridge demonstrates that thoughtfully applied artificial intelligence, economic mechanism design (token economy), and established behavioural economics principles (blind reviews) can — in combination — substantially improve the fairness, efficiency, and trust of digital freelance marketplaces. This project is submitted with confidence that it meets and exceeds the expectations set in the original FYP proposal.

Appendix A

API Reference

This appendix provides a concise reference for the primary SkillBridge REST API endpoints. All endpoints are prefixed with `/api`. Protected endpoints require a valid `Authorization: Bearer <access_token>` header unless otherwise noted.

A.1 Authentication Routes (`/api/auth`)

Method	Endpoint	Auth	Description
POST	/signup	Public	Register a new user account
POST	/verify-otp	Public	Verify email OTP after signup
POST	/login	Public	Authenticate and receive access token
POST	/logout	Public	Invalidate the refresh token session
POST	/refresh	Public	Obtain a new access token via cookie refresh
POST	/forgot-password	Public	Send password reset email
POST	/reset-password	Public	Reset password using token
GET	/google	Public	Initiate Google OAuth 2.0 flow

Method	Endpoint	Auth	Description
GET	/google/callback	Public	Google OAuth callback handler
POST	/complete-google-signup	Protected	Complete role selection after Google OAuth

A.2 Project Routes (/api/projects)

Method	Endpoint	Auth	Description
POST	/	CLIENT	Create a new project (supports file upload)
PATCH	/:id	CLIENT	Update an existing project
GET	/	Protected	List all open projects
GET	/client/my	CLIENT	List authenticated client's projects
GET	/:id	Protected	Get project details by ID
DELETE	/:id	CLIENT	Delete a project

A.3 Proposal Routes (/api/proposals)

Method	Endpoint	Auth	Description
POST	/	FREELANCER	Submit a proposal (deducts SkillTokens)
GET	/project/:projectId	CLIENT	List all proposals for a project
GET	/my	FREELANCER	List authenticated freelancer's proposals
PATCH	/:id/withdraw	FREELANCER	Withdraw a proposal (partial token refund)

Method	Endpoint	Auth	Description
PATCH	/:id/shortlist	CLIENT	Shortlist a proposal
PATCH	/:id/accept	CLIENT	Accept a proposal (creates Contract)
PATCH	/:id/reject	CLIENT	Reject a proposal

A.4 Token Routes (/api/tokens)

Method	Endpoint	Auth	Description
GET	/balance	FREELANCER	Get current SkillToken balance
GET	/history	FREELANCER	Get full transaction history
POST	/admin/grant	ADMIN	Admin grant tokens to a freelancer
POST	/admin/deduct	ADMIN	Admin deduct tokens from a freelancer

A.5 Freelancer Profile Routes (/api/freelancers)

Method	Endpoint	Auth	Description
GET	/me	FREELANCER	Get authenticated freelancer's profile
PATCH	/me	FREELANCER	Update authenticated freelancer's profile
GET	/:id	Protected	Get a freelancer's public profile by ID
POST	/me/skills	FREELANCER	Add skills to profile
DELETE	/me/skills/:skillId	FREELANCER	Remove a skill from profile
POST	/me/portfolio	FREELANCER	Add a portfolio item

Method	Endpoint	Auth	Description
DELETE	/me/portfolio/:id	FREELANCER	Remove a portfolio item
POST	/me/gigs	FREELANCER	Upload a gig document
DELETE	/me/gigs/:id	FREELANCER	Remove a gig

A.6 AI Scoping Route (`/api/ai`)

Method	Endpoint	Auth	Description
POST	/scope	CLIENT	Submit a natural language project description; returns structured scoping JSON from Gemini

A.7 Prisma Schema Entity Summary

Table A.7: *SkillBridge Prisma Schema Models*

Model	**Purpose**
User	Unified auth record for all roles
Session	JWT refresh token tracking per device
ClientProfile	Extended profile for CLIENT users
FreelancerProfile	Extended profile for FREELANCER users
AdminProfile	Extended profile for ADMIN users
Skill	Master list of available skills
FreelancerSkill	Joins FreelancerProfile to Skill with proficiency level
PortfolioItem	Freelancer past work showcase items
Certificate	Freelancer credentials and certifications
Education	Freelancer education records
Gig	Freelancer pre-packaged service documents
Category	Project top-level category
SubCategory	Project sub-category under a Category
Project	Core marketplace project entity
ProjectSkill	Joins Project to Skill
AiScopingSession	Stores AI scoping request/response pairs
Proposal	Freelancer bid on a project
TokenTransaction	Immutable SkillToken ledger entries
Invitation	Client invitation to freelancer
Contract	Agreement upon proposal acceptance
Milestone	Deliverable checkpoint within a contract
Payment	Escrow and release record per milestone
ChatRoom	Message thread per contract/inquiry
Message	Chat message within a ChatRoom
Review	Blind review post-contract
Dispute	Disagreement record managed by Admin
Notification	Per-user activity notification feed
AdminLog	Audit trail for admin actions
PlatformSetting	Admin-controlled key-value settings store
PasswordResetToken	Time-limited token for password reset
SavedProject	Freelancer bookmarked projects
BrowseInteraction	Browse system interaction logs
Language	Language reference data (metadata)
Location	Location reference data (metadata)

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